

Transformer Engineering Data Manual

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1. Transformer Full Load Currents (kVA to Amperes)

Rating kVA	400V 3Ph (A)	415V 3Ph (A)	11kV 3Ph (A)	22kV 3Ph (A)	%Z Impedance
25 kVA	36.08 A	34.78 A	1.31 A	0.66 A	4.0%
50 kVA	72.17 A	69.56 A	2.62 A	1.31 A	4.0%
100 kVA	144.34 A	139.12 A	5.25 A	2.62 A	4.0%
250 kVA	360.84 A	347.80 A	13.12 A	6.56 A	4.0%
500 kVA	721.69 A	695.60 A	26.24 A	13.12 A	4.5%
800 kVA	1154.70 A	1112.96 A	41.99 A	20.99 A	5.0%
1000 kVA	1443.38 A	1391.20 A	52.49 A	26.24 A	5.0%
1600 kVA	2309.40 A	2225.93 A	83.98 A	41.99 A	6.0%
2000 kVA	2886.75 A	2782.41 A	104.97 A	52.49 A	6.0%
2500 kVA	3608.44 A	3478.01 A	131.22 A	65.61 A	6.5%

2. Short-Circuit Currents and Impedance Calculation

Prospective fault current: $I_{sc} = I_{rated} / (\%Z / 100)$

Example: 1000 kVA at 400V, %Z = 5.0% -> $I_{sc} = 1443.4 / 0.05 = 28.87$ kA.

Secondary main circuit breaker must have breaking capacity ≥ 36 kA.

3. Vector Groups (Dyn11, Dyn5, Yyn0)

Dyn11: Delta HV, Star Neutral LV, LV leads HV by 30 deg (standard distribution).

Dyn5: Delta HV, Star Neutral LV, LV lags HV by 150 deg.

Yyn0: Star HV, Star Neutral LV, 0 deg phase displacement.